

Tech Lab Equipment Procurement Guide - Grades 6-8

Lab 1: Computer Architecture & Hardware Systems Lab

Equipment List & Procurement:

Core Hardware Components:

Equipment	Quantity	Supplier	Estimated Price Range
Transparent Desktop PC Kits	8 sets	Amazon India, Robu.in	₹12,500-16,700 each
Raspberry Pi 5 Starter Kits	16 units	RS Components India, Robu.in	₹6,700-10,000 each
Arduino Uno R4 Development Boards	20 units	Arduino Store, Robu.in	₹2,100-2,900 each
ESP32-S3 Development Boards	16 units	Robu.in, TechnoBotOnline	₹1,250-2,100 each
Digital Storage Oscilloscopes (2-channel)	4 units	Rigol India, Test Equipment Store	₹33,400-66,700 each
Logic Analyzers	4 units	Saleae, Hantek India	₹16,700-41,700 each
EEPROM Programmers	2 units	Xeltek, TL866 India	₹8,350-16,700 each
Anti-static Workstations	8 stations	StarTech India, ESD Solutions	₹16,700-25,000 each
Component Testers	4 units	ESR meters, LCR meters	₹4,200-12,500 each

Lab Infrastructure:

Equipment	Quantity	Supplier	Estimated Price Range
Adjustable Lab Tables	8 units	Lab furniture suppliers India	₹25,000-41,700 each

Tool Storage Cabinets	4 units	Uline India, Industrial suppliers	₹16,700-33,400 each
Power Distribution Units	4 units	APC India, CyberPower	₹8,350-16,700 each
ESD Floor Mats	1 set	Desco India, ESD Solutions	₹41,700-66,700 total

Lab 1 Total Investment: ₹15,00,000 - ₹23,30,000

Lab 2: AI & Computational Intelligence Lab

Equipment List & Procurement:

Computing Hardware:

Equipment	Quantity	Supplier	Estimated Price Range
Student Laptops/Tablets	16 units	Dell India, HP India, Apple India	₹33,400-66,700 each
NVIDIA Jetson Nano Dev Kits	8 units	NVIDIA India, Arrow Electronics	₹8,350-12,500 each
AI Cameras (USB 3.0)	12 units	Logitech India, Intel India	₹6,700-16,700 each
Smart Sensor Kits	8 sets	Grove sensors, DFRobot India	₹12,500-20,900 each
Programmable Robots	8 units	Sphero India, micro:bit India	₹8,350-25,000 each

Software & Platforms:

Software/Service	Quantity	Supplier	Estimated Price Range
Educational Software Licenses	16 licenses	Microsoft India, Google India	₹4,200-8,350 per student/year
Cloud Computing Credits	Annual	AWS India, Google Cloud India	₹83,500-1,67,000/year

AI Development Platforms	Site license	Various Indian distributors	₹41,700-1,25,000/year
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Lab Infrastructure:

Equipment	Quantity	Supplier	Estimated Price Range
Interactive Whiteboards	2 units	SMART India, Promethean India	₹1,25,000-2,50,000 each
High-Speed Internet Router	1 unit	Cisco India, Ubiquiti India	₹25,000-50,000
Network Storage (NAS)	1 unit	Synology India, QNAP India	₹41,700-83,500

Lab 2 Total Investment: ₹12,50,000 - ₹20,80,000

Lab 3: Advanced Electronics & Embedded Systems Lab

Equipment List & Procurement:

Electronics Development:

Equipment	Quantity	Supplier	Estimated Price Range
Soldering Stations with Fume Extraction	8 units	Weller India, Hakko India	₹16,700-33,400 each
Digital Multimeters	16 units	Fluke India, Keysight India	₹8,350-25,000 each
DC Power Supplies (Variable)	8 units	Tekpower India, BK Precision	₹12,500-25,000 each
Function Generators	4 units	Siglent India, Rigol India	₹25,000-50,000 each
Advanced Breadboarding Stations	16 sets	Global Specialties India	₹4,200-8,350 each

Arduino Ecosystem Kits	16 sets	Arduino India, Elegoo India	₹6,700-12,500 each
Sensor Collections	8 sets	Grove India, Adafruit India	₹16,700-33,400 each

PCB Development:

Equipment	Quantity	Supplier	Estimated Price Range
PCB Design Software Licenses	16 licenses	Autodesk India, KiCad Pro	₹8,350-25,000 per license
PCB Prototyping Machine	1 unit	LPKF India, Bantam Tools	₹12,50,000-20,80,000
Hot Air Rework Stations	4 units	Quick India, Aoyue India	₹16,700-41,700 each

Lab 3 Total Investment: ₹20,80,000 - ₹33,30,000

Lab 4: Advanced Making & Fabrication Lab

Equipment List & Procurement:

Digital Fabrication:

Equipment	Quantity	Supplier	Estimated Price Range
Multi-material 3D Printers	4 units	Prusa India, Ultimaker India	₹66,700-2,08,000 each
Laser Cutter/Engraver	1 unit	Glowforge India, Epilog India	₹3,33,000-12,50,000
Vinyl Cutter	2 units	Cricut India, Silhouette India	₹16,700-41,700 each
CNC Router (Desktop)	1 unit	Shapeoko India, X-Carve India	₹1,67,000-4,17,000

3D Scanner	1 unit	Creality India, Revopoint India	₹25,000-1,25,000
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Traditional Workshop Tools:

Equipment	Quantity	Supplier	Estimated Price Range
Workbenches with Vises	8 units	Sjobergs India, Industrial suppliers	₹25,000-66,700 each
Hand Tool Sets	8 sets	Stanley India, Craftsman India	₹12,500-25,000 each
Drill Press	1 unit	Delta India, JET India	₹25,000-66,700
Band Saw (Small)	1 unit	Porter-Cable India, Rikon India	₹33,400-83,500
Safety Equipment	Complete set	3M India, Honeywell India	₹41,700-83,500

Design Software:

Software	Quantity	Supplier	Estimated Price Range
Fusion 360 Educational	16 licenses	Autodesk India	Free for education
Adobe Creative Suite	16 licenses	Adobe India	₹20,000/year per license

Lab 4 Total Investment: ₹16,70,000 - ₹29,20,000

Lab 5: Systems Integration & Robotics Lab

Equipment List & Procurement:

Robotics Platforms:

Equipment	Quantity	Supplier	Estimated Price Range
VEX IQ Robotics Kits	8 sets	VEX Robotics India	₹25,000-41,700 each

LEGO SPIKE Prime Sets	8 sets	LEGO Education India	₹27,500-33,400 each
Custom Arduino Robot Chassis	16 kits	Custom assembly India	₹8,350-16,700 each
Educational Drone Kits	4 units	DJI Tello EDU India, Parrot India	₹8,350-25,000 each
6-DOF Robotic Arms	2 units	Dobot India, uArm India	₹66,700-1,67,000 each

Control & Programming:

Equipment	Quantity	Supplier	Estimated Price Range
Competition Mats & Fields	2 sets	VEX India, FIRST India	₹41,700-1,25,000 each
Programming Tablets	16 units	iPad India, Android tablets	₹16,700-41,700 each
Wireless Router (Dedicated)	1 unit	Cisco India, Ubiquiti India	₹16,700-33,400

Lab 5 Total Investment: ₹12,50,000 - ₹20,80,000

Lab 6: Future Technology & Innovation Lab

Equipment List & Procurement:

Emerging Technology:

Equipment	Quantity	Supplier	Estimated Price Range
VR/AR Headsets	8 units	Meta Quest India, HoloLens India	₹25,000-1,25,000 each
Quantum Computing Simulator	1 kit	IBM Qiskit India, educational suppliers	₹41,700-1,67,000

Biomaker Equipment Set	1 set	pH meters, microscopes India	₹1,67,000-4,17,000
Renewable Energy Demo Kits	4 sets	Vernier India, PASCO India	₹25,000-66,700 each
Smart Materials Samples	1 collection	Educational suppliers India	₹41,700-1,25,000

Innovation Tools:

Equipment	Quantity	Supplier	Estimated Price Range
Interactive Display	1 unit	Microsoft Surface Hub India	₹2,50,000-6,67,000
Video Production Kit	1 set	Canon India, Sony India	₹1,25,000-2,50,000
Rapid Prototyping Materials	Ongoing	Various suppliers India	₹41,700-83,500/year
Digital Collaboration Tools	Site license	Microsoft India, Google India	₹83,500-2,50,000/year

Lab 6 Total Investment: ₹10,00,000 - ₹20,80,000

Recommended Suppliers by Category

Electronics & Components:

- **Robu.in** (robu.in) - Leading Indian electronics supplier
- **TechnoBotOnline** (technobotsonline.com) - Arduino & IoT components
- **Rhydo Technologies** (rhydolabz.com) - Educational electronics
- **Amazon India Business** (business.amazon.in) - General equipment
- **RS Components India** (in.rs-online.com) - Professional components

Test Equipment:

- **Tektronix India** (tek.com/india) - Professional oscilloscopes
- **Rigol India** (rigol.co.in) - Affordable test equipment

- **Keysight India** (keysight.com/in) - High-end measurement tools
- **Scientific Mes-Technik** (scientificmesttechnik.com) - Logic analyzers
- **Fluke India** (fluke.com/en-in) - Quality multimeters

Fabrication Equipment:

- **Prusa Research India** (prusa3d.com) - 3D printers
- **Think3D** (think3d.in) - Indian 3D printing solutions
- **Inventor Electronics** (inventorelectronics.com) - CNC machines
- **Industrial Tool House** - Workshop tools
- **Amazon Industrial India** - General fabrication supplies

Educational Technology:

- **VEX Robotics India** (vexrobotics.com/india) - Robotics platforms
- **LEGO Education India** (education.lego.com/en-in) - Educational robotics
- **Sphero India** (sphero.com) - Programming robots
- **Microsoft India Education** (microsoft.com/en-in/education) - Software licenses
- **Google for Education India** (edu.google.com) - Cloud platforms

Lab Furniture & Infrastructure:

- **Lab Equipment India** - Laboratory furniture
- **Industrial Furniture Suppliers** - Storage solutions
- **Godrej Interio** (godrejinterio.com) - Workbenches
- **IndiaMART** (indiamart.com) - Industrial supplies marketplace

Total Investment Summary

Lab	Low Estimate	High Estimate
Lab 1: Computer Architecture	₹15,00,000	₹23,30,000

Lab 2: AI & Computational Intelligence	₹12,50,000	₹20,80,000
Lab 3: Advanced Electronics	₹20,80,000	₹33,30,000
Lab 4: Advanced Making & Fabrication	₹16,70,000	₹29,20,000
Lab 5: Systems Integration & Robotics	₹12,50,000	₹20,80,000
Lab 6: Future Technology & Innovation	₹10,00,000	₹20,80,000

TOTAL INITIAL INVESTMENT: ₹87,50,000 - ₹1,48,20,000

Annual Operating Costs:

- Software licenses and subscriptions: ₹4,17,000 - ₹8,33,000
- Consumables and materials: ₹6,67,000 - ₹12,50,000
- Equipment maintenance: ₹2,50,000 - ₹5,00,000
- Professional development: ₹4,17,000 - ₹8,33,000

TOTAL ANNUAL OPERATING: ₹17,51,000 - ₹34,16,000

Implementation Strategy

Phase 1 (Year 1): Core Labs Setup

- Priority: Labs 1, 2, and 3 (Essential foundation)
- Investment: ₹48,30,000 - ₹77,40,000

Phase 2 (Year 2): Advanced Capabilities

- Add: Labs 4 and 5 (Making and Robotics)
- Additional Investment: ₹29,20,000 - ₹50,00,000

Phase 3 (Year 3): Innovation Completion

- Add: Lab 6 (Future Technology)
- Final Investment: ₹10,00,000 - ₹20,80,000

Funding Recommendations:

- **Government Grants:** Digital India, Skill India initiatives
- **Corporate Partnerships:** TCS, Infosys, Wipro CSR programs

- **Educational Foundations:** Azim Premji Foundation, equipment donations
- **Parent/Community Support:** Crowdfunding, local business sponsorship
- **Phased Implementation:** Spread costs over 2-3 years

Note: Prices are estimates based on current Indian market rates (₹83.50 = \$1 USD) and may vary based on quantity discounts, educational pricing, and market fluctuations. Always request quotes from multiple Indian suppliers for accurate budgeting.

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